

The Future of Human and Machine

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Lecture for Faculty of ICT, Mahidol University

Overview

- Background
- Technology Trends for Computer Engineering Research
- Reality Inside and Outside the Screen – Mixed/Augmented Reality

<http://gscm.nida.ac.th/~chutisant/MU.pdf>

Education



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp/

- MSc. / Ph.D. in Engineering, Department of
Information and Computer Science
Keio University, the oldest university in Japan

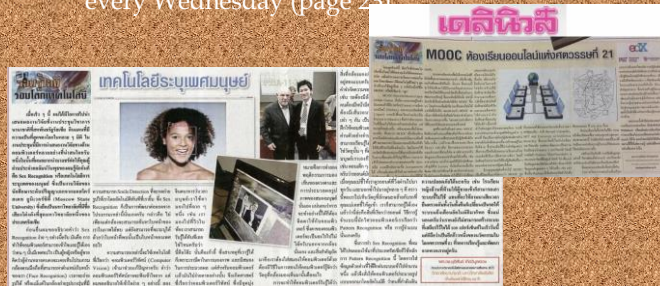
Academic Work Experience

- Teaching and being as a research committee in following universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Kasetsart University, NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

Media Work Experience (1/3)

1. Newspapers

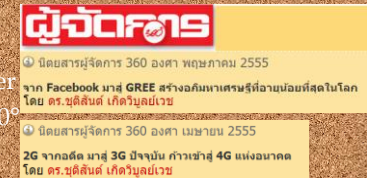
- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



Media Work Experience (2/3)

2. Magazines

- Columnist for Manager 360° Magazine, iTech 360° Magazine, Augmented Reality Magazine, e-commerce Magazine



Media Work Experience (3/3)

3. Televisions

- IT News Analyst for Nation Channel TV Program



- IT News Analyst for RSU TV Channel



TECHNOLOGY TRENDS FOR COMPUTER ENGINEERING RESEARCH

Digital Media

Digital media are any media that are encoded in a machine-readable format. [Technology Brief. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



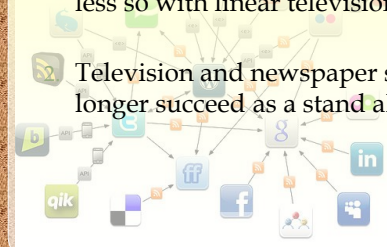
www.linkedin.com

Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.



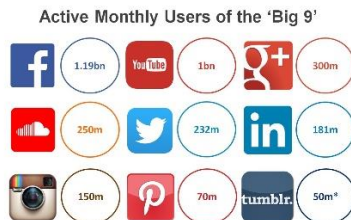
Television and newspaper still matter, but it can no longer succeed as a stand alone platform.

Source:
Digital Media Trends and the Future of
Television, Dr. Seth Geiger,
President SmithGeiger LLC (CA, USA)

<https://www.linkedin.com/>

Trends for Digital Media Trends for 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
*Tumblr does not publish active monthly user data and this figure is based on an estimate from
Twitter and other sources. Twitter and other sources do not publish active monthly user data.

kamber

Source:
Kamber (Sydney,
Australia)

kamber

14 Social Media Trends for 2014

Kamber.com.au 14

Mobile Phone As Home Computer

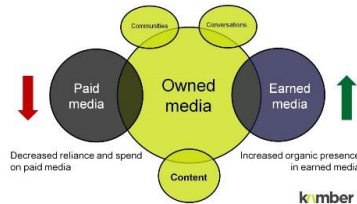
- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



voices.suntimes.com

Paid Media / Earned Media

The content marketing argument



::Source::
Kamber (Sydney,
Australia)

kimber 14 Social Media Trends for 2014 Kimber.com.au 10

Creative Digital Economy

Placement: Its where young (and old) people are

Price to invest: Minimal

But...

- Easy to build innovatively
- Create HUGE profits



<http://www.dailynews.co.th/>

What's Next?

- ▣ Big Data / Cloud Computing / SEO (Search Engine Optimization) - SEM (Search Engine Marketing)
- ▣ Mobile-Oriented
- ▣ Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



www.3dsg.com.au

3D DISPLAY / 3D TELEVISION

3D Display / 3D Television



<http://www.afp.com/>

3D Display / 3D Television

A free-floating image created by firing lasers into thin air

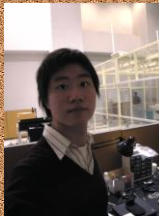


<http://www.afp.com/>

3D Display / 3D Television

- ❑ 2008-Current
- ❑ Hideki Kimura
- ❑ National Museum of Emerging Science and Innovation (Miraikan), Tokyo

2009



2014 October



<http://www.afp.com/>

3D Display / 3D Television



<http://www.afp.com/>

3D Display / 3D Television

- ❑ Miraikan is located in Odaiba, Tokyo



www.japan-guide.com

NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,
Carnegie Mellon University (CMU),
United States

Super Bowl

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

Super Bowl

- ❑ “Eye Vision”
- ❑ Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- ❑ The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



Eye Vision from Prof.
Takeo Kanade, CMU

The Matrix



- ❑ "Virtualized Reality" www.imdb.com
- ❑ People will be able to customize the perspective from which they watch, for example, from that of a particular object.

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New Viewpoint Analysis

AUGMENTED REALITY

What is Augmented Reality (AR)?



- ❑ A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- ❑ Generates a composite view for the user.

What is the Goal of AR?

- ❑ To enhance a person's performance and perception of the world
- ❑ But, what is the ultimate



'Augmented reality in urban places: contested content and the duplicity of code.' Transactions of the Institute of British Geographers, 2012

The Ultimate Goal of AR

- ❑ Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce Geek.com

Augmented Reality vs. Virtual Reality

Augmented Reality

- ❑ System augments the real world scene
- ❑ User maintains a sense of presence in real world
- ❑ Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- ❑ Totally immersive environment
- ❑ Visual senses are under control of system

Miligram's Reality-Virtuality Continuum



Miligram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

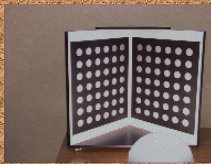
Combining the Real and Virtual Worlds

We need:

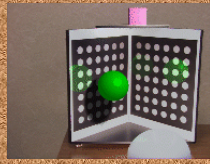
- ❑ Precise models
- ❑ Locations and optical properties of the viewer (or camera) and the display
- ❑ Calibration of all devices
- ❑ To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

Combining the Real and Virtual Worlds (cont)

- ❑ Register models of all 3D objects of interest with their counterparts in the scene
- ❑ Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-reflections and virtual shading

Mixed Reality

- ❑ Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing



Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

This image depicts an artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sonycsi.co.jp

Realistic Merging

Requires:

- ❑ Objects to behave in physically plausible manners when manipulated
- ❑ Occlusion
- ❑ Collision detection
- ❑ Shadows

**All of this requires a very detailed description of the physical scene

Research Activities

- ❑ Develop methods to register the two distinct sets of images and keep them registered in real-time
 - New work in this area has started to use Computer Vision (CV) techniques
- ❑ Develop new display technologies for merging the two images

Performance Issues

Augmented Reality systems are expected:

- ❑ To run in real-time so that the user can move around freely in the environment
- ❑ Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- ❑ Update rate for generating the augmenting image
- ❑ Accuracy of the registration of the real and virtual image

Failures in Registration

Failures in registration due to:

- Noise
 - Position and pose of camera with respect to the real scene
 - Fluctuations of values while the system is running
- Time delays
 - In calculating the camera position
 - In calculating the correct alignment of the graphics camera

Applications

- ❑ Medical
- ❑ Military Training
- ❑ Hazard Detection
- ❑ Engineering Design
- ❑ Robotics and Telerobotics
- ❑ Training
- ❑ Sports
- ❑ Tourism
- ❑ Video conferencing

Applications (cont.)

- ❑ Entertainment, Computer Games
- ❑ Manufacturing, Maintenance, and Repair
- ❑ Consumer Design
- ❑ Audio
- ❑ Education
- ❑ Architecture
- ❑ Human welfare

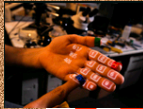
RECENT AUGMENTED REALITY APPLICATIONS

SIXTH SENSE

Sixth Sense

- ❑ A wearable gestural interface that augments the physical world around us with digital information
- ❑ Let us use natural hand gestures to interact with that information
- ❑ The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Sixth Sense in 21st Century



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED



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Detail of Sixth Sense

- ❑ Sixth Sense allows people to use the internet without a screen or a keyboard
- ❑ It can turn any surface into a touch screen
- ❑ It is made with store-bought components
- ❑ *"The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques"* ...from MIT Media Lab

Detail of Sixth Sense

- ❑ The device recognizes items in stores, and then gathers and projects information about products to the user.
- ❑ It also is able to provide quick signals to let users know what the best products are based on their needs."
- ❑ Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- ❑ Say goodbye to your encyclopedia

Cost of Sixth Sense

- ❑ Only \$350 dollars, around 12,000 baht
- ❑ From an educator's stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- ❑ For educator's, Sixth Sense would be a way to put technology (literally) into each student's hands, because of its size and its cost.

AUGMENTED REALITY GUITAR

Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel



MR FOR MILITARY APPLICATION

Mixed Reality for Military Application



Dr. Adrian Cheok,
National University of Singapore (NUS)

Broadcast in CNN



AUGMENTED REALITY MAGIC

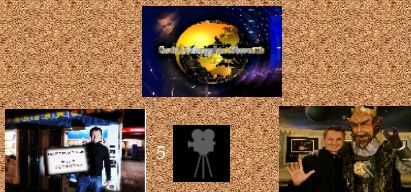
Augmented Reality Magic

- ❑ Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- ❑ Along comes a pasteboard prestidigitator who folds augmented reality into



Augmented Reality Magic (card trick)

- ❑ Blend of technology and illusion creates the magic of tomorrow, today



Marco Tempest,
Newmagic Communications
Inc.,
United States

Integration leads to Innovation

Traditional Methods



Conventional
Frameworks

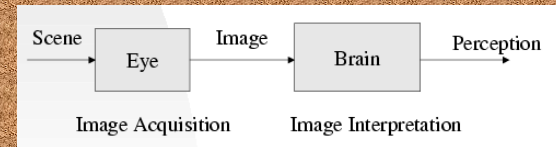
Research
Innovation

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INTRODUCTION TO COMPUTER VISION

Vision

- ▣ Vision is the process of discovering what is present in the world and where it is by looking.

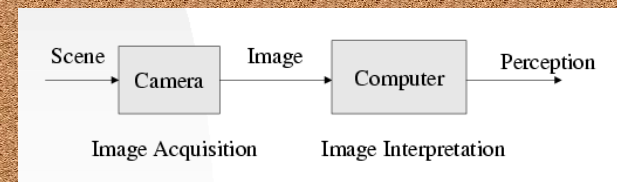


Computer Vision

- ▣ “**Computer vision** is the science and technology of machines that see. As a scientific discipline, computer vision is concerned with the theory behind artificial systems that extract information from images. The image data can take many forms, such as video sequences, views from multiple cameras, or multi-dimensional data from a medical scanner.”

Computer Vision

- ▣ Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



Introduction of Computer Vision

□ Digital Image Processing (IP)

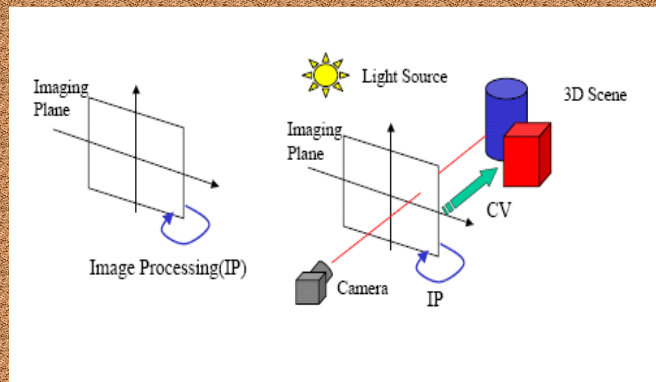
- Since 1960's
- 1970's : Recognized as a technology for practical use.
- Most Important Issues : Image Transformation
- Noise/Blur Compensation, Distortion Correction

Introduction of Computer Vision

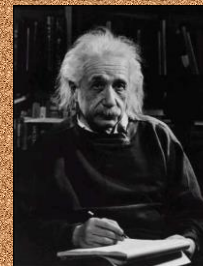
□ Computer Vision (CV)

- Emerged from AI (Artificial Intelligence) Research
- Computational Vision proposed by Marr
 - Shape from X(X:shading, texture, contour, motion...etc.)
 - 3D Scene Understanding from 2D images
- Robot Vision: Vision functionality for Robots
- Media Processing, Human Interface

Image Processing / Computer Vision



Computer Vision



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Components of a Computer Vision System

